

THE VERIFIER BOTTLENECK

Can a model combine two skills after we teach them separately?



A ⊕ B

`compose · solve · verify`

WHAT DO WE CHANGE IN THE EXPERIMENT?

VERIFIER SIGNAL

WHICH ANSWERS GO BACK INTO TRAINING?

$\alpha = P(\text{accept} \mid \text{correct})$

$\beta = P(\text{accept} \mid \text{wrong})$

EXPLORATION

HOW MANY ATTEMPTS DO WE GIVE THE MODEL?

candidate budget K

temperature and diversity

SHARPENING

``pass@1`` rises; fixed-budget ``pass@k`` stays flat.

OPERATIONAL EXPANSION

Held-out fixed-budget ``pass@k`` rises.

COLLAPSE

Held-out fixed-budget ``pass@k`` falls.

TOY TASK: LEARN A AND B, THEN SOLVE AB.

A**COMPOSE** $g_3(g_2(g_1(x)))$ $\rightarrow A \cdot x + B \pmod{p}$

Collapse an ordered chain into
one affine map.

+

B**INVERT** $A \cdot x + B = y$ $\rightarrow x = A^{-1}(y-B)$

Recover x with modular inversion.

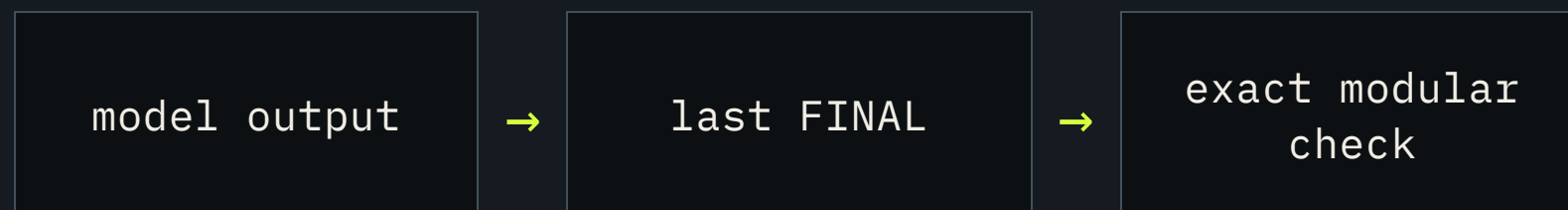
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AB**COMPOSE AND SOLVE** $g_3(g_2(g_1(x))) = y$ $\rightarrow$ find x

Collapse-first and backwards
inversion both count.

THE ORACLE CHECKS TRUTH. NOISE CHANGES ONLY THE REWARD.

EVALUATION LANE PERFECT ORACLE



Fresh test prompts are always scored here. Noise never changes the gold answer.

TRAINING LANE • OPTIONAL FOLLOW-UP NOISY REWARD

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u = H(event_id) / 264  
accept if u < α when correct  
accept if u < β when wrong
```

The key contains the rollout ID, not the answer. Retrying one rollout gives the same draw; a new rollout gets a new draw.

FIRST RUN: REJECTION SFT WITH LORA.

01 ATOMIC LORA SFT Teach A and B separately. No AB solutions.	02 FREEZE MO Every condition starts from the same atomic checkpoint.	03 SAVE ONE POOL 64 AB attempts per prompt. K=8 is a prefix of that pool.	04 FILTER AND UPDATE No update, exact filter, and rate-matched random filter. One rejection- SFT/LoRA update.	05 FRESH EVALUATION AB pass@1/pass@16 plus A/B retention. GRPO WAITS
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WHICH MODEL SHOULD WE START WITH?

FIRST CANDIDATE

Qwen2.5-Math-
1.5B-Instruct

Use it only if A and B are learned and
AB is not already saturated.

IF THE TASK IS TOO EASY

weaker checkpoint
or deeper chains

A base model that already solves every
AB task leaves nothing to test.

SECOND MODEL LATER

Llama-3.2-
3B-Instruct

Add it only after the first comparison
is clear enough to interpret.

FREEZE BEFORE COMPARISON

checkpoint · chat template · seeds · response limits · decoding · LoRA settings · optimizer steps · generated-token budget

BEFORE THE FIRST RUN.

01 Approve the A/B/AB task and the held-out split.

02 Choose the first checkpoint and available hardware budget.

03 Agree on the pilot calibration and stop gate.

04 Freeze the manifest, decoding settings, and update budget.

ALREADY IMPLEMENTED

- ✓ public/private toy rows
- ✓ exact modular checker
- ✓ deterministic noisy wrapper
- ✓ experiment plan

We have not trained a model yet. This deck describes the proposed experiment.